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Paper 1: A simple algebraic expression can determine if a drug's clearance is non-specific

Bibliographic Information

- **Title:** A simple algebraic expression can determine if a drug's clearance is non-specific or target-mediated
- **Authors:** Brown LV, Ippolito A, Penney M, et al.
- **Journal:** Journal of Pharmacokinetics and Pharmacodynamics, 2026, 53:39
- **DOI:** 10.1007/s10928-026-10043-w

Paper Type: METHODOLOGY

Quick Take

Brown et al. derive a simple algebraic expression to determine whether a biologic's clearance is dominated by non-specific (intrinsic) routes or by target-mediated drug disposition (TMDD). The expression depends only on the intrinsic clearance rate, target turnover rate, binding affinity, and receptor concentration. Applied to T-cell engagers and monoclonal antibodies, it correctly predicts that TCEs are TMDD-driven at clinical concentrations while most mAbs are intrinsic-driven, providing a practical rule-of-thumb for cross-species PK scaling.

Executive Summary

Brown et al. derive a simple algebraic expression for the proportion of a biologic's clearance that is due to non-specific (intrinsic) routes versus target-mediated drug disposition (TMDD). The expression depends only on the intrinsic clearance rate, the target receptor's turnover rate, its binding affinity, and the free ligand and receptor concentrations. They show that the transition from intrinsic-dominated to TMDD-dominated clearance occurs at a ligand concentration given by $(k_e/k_{clear}) \cdot R_{tot} - k'_D$, and that when this quantity is negative, TMDD never dominates. The authors validate their expressions against published PK data for several T-cell engagers and monoclonal antibodies, demonstrating that TCEs are consistently predicted to have TMDD-driven clearance at pharmacologically active concentrations, while most mAbs are intrinsic-driven. This work provides a practical rule-of-thumb for determining whether allometric scaling (for intrinsic clearance) or target-parameter-based scaling (for TMDD) should be used for cross-species PK translation.

Scientific Context & Motivation

Target-mediated drug disposition (TMDD) is a well-known phenomenon where a drug's clearance is accelerated at low concentrations due to binding and internalization with its target. For monoclonal antibodies, TMDD is typically 'out-dosed' by giving high doses that saturate the target, resulting in monoexponential PK driven by intrinsic (non-specific) clearance. However, T-cell

engagers (TCEs) and other immune engagers are often dosed at low levels to avoid toxicity, yet their PK often appears monoexponential. This creates a dilemma: is the observed monoexponential clearance due to intrinsic routes (which would scale allometrically between species) or due to TMDD (which would not)? The authors address this gap by deriving a simple algebraic expression that quantifies the proportion of clearance that is intrinsic vs. TMDD, using only readily available parameters (k_{clear} , k_e , R_{tot} , k_D , k_{on}). This allows DMPK scientists to quickly determine which clearance mechanism dominates and thus which scaling method is appropriate for cross-species translation.

✂ Methodological Snapshot

The authors derive an algebraic expression for the proportion of a biologic's clearance that is due to non-specific (intrinsic) routes versus target-mediated drug disposition (TMDD). Starting from the standard TMDD ODE system under the quasi-steady-state (QSS) approximation, they rewrite the total ligand derivative to isolate the intrinsic and TMDD contributions. This yields $p_{\text{intrinsic}} = 1/(1 + (k_e/k_{\text{clear}})*(R_{\text{free}}/k'_D))$, where k_{clear} is the intrinsic clearance rate, k_e is the complex elimination rate, R_{free} is the free receptor concentration, and $k'_D = k_D + k_e/k_{\text{on}}$ is the QSS dissociation constant. The threshold for 50% intrinsic clearance is $L_{\text{free}} = (k_e/k_{\text{clear}})*R_{\text{tot}} - k'_D$. The authors validate these expressions through numerical simulations and application to published PK data for T-cell engagers and monoclonal antibodies. They also extend the analysis to two-compartment models and trimeric (bispecific) binding in the supplementary materials.

⊠ Statistical Framework

The work is based on the standard TMDD ODE model (Mager & Jusko, 2001) under the quasi-steady-state (QSS) approximation (Gibiansky et al., 2008). The key assumption is that binding kinetics (k_{on} , k_{off}) are fast relative to other processes (clearance, turnover), so that the complex concentration C can be considered in steady state with respect to free ligand and receptor. This yields the algebraic relationship $C = (L_{\text{free}} * R_{\text{free}}) / k'_D$, where $k'_D = (k_{\text{off}} + k_e)/k_{\text{on}} = k_D + k_e/k_{\text{on}}$. The authors then derive the total ligand derivative $dL_{\text{tot}}/dt = -(k_{\text{clear}} + k_e * R_{\text{free}}/k'_D) * L_{\text{free}}$, from which the proportion of intrinsic clearance follows directly. The derivation is deterministic and exact under the QSS assumption. No stochastic or statistical inference is performed; the validation is through numerical simulation and comparison to published PK data. The authors also consider the case of finite k_{on} (Appendix A.2) and show that the QSS approximation may break down at very low L_{free} , but the derived thresholds remain useful as rules-of-thumb.

Estimator Behavior

The derived expressions are not estimators in the statistical sense but rather algebraic identities that hold exactly under the quasi-steady-state (QSS) approximation. The proportion of intrinsic clearance $p_{\text{intrinsic}} = 1/(1 + (k_e/k_{\text{clear}})*(R_{\text{free}}/k'_D))$ is a deterministic function of the state

variables and parameters. Under the QSS assumption, it is exact. The threshold $L_{\text{free}} = (k_e/k_{\text{clear}}) \cdot R_{\text{tot}} - k'_D$ defines the concentration at which $p_{\text{intrinsic}} = 0.5$. The authors show that when L_{free} is far above this threshold, clearance is dominated by intrinsic routes (monoexponential decay with rate k_{clear}), and when L_{free} is far below it, clearance is dominated by TMDD (monoexponential decay with rate $k_e \cdot R_{\text{tot}} / (k'_D + R_{\text{tot}})$). The expressions are unbiased under the QSS assumption but may be biased if binding is not fast relative to other processes (finite k_{on}). The authors demonstrate this in Appendix A.2 (Figure 7) where finite k_{on} causes divergence between theoretical and numerical $p_{\text{intrinsic}}$ at low L_{free} .

Validation Design

The authors validate their derived expressions through two approaches. First, numerical simulations of the full TMDD ODE system (Eq. 1) are compared to the theoretical predictions. In Figure 3, simulated free ligand concentration vs. time curves are annotated with the derived thresholds (Eq. 7 and $L_{\text{free}} = k_D$), showing that the thresholds correctly predict the onset of TMDD-driven monoexponential decay and the ‘wobble’ region. The simulations cover a wide range of parameter values (k_D , R_{tot} , initial dose). Second, the expressions are applied to published clinical PK data for 5 T-cell engagers (Epcoritamab, Glofitamab, Talquetamab, Teclistamab, Mosunetuzumab) and 7 monoclonal antibodies (Anifrolumab, Canakinumab, Sifalimumab, Sirukumab, Brodalumab, Tocilizumab, AMG108). For each compound, literature values for target parameters (R_{tot} , k_D , k_{on} , k_e) and intrinsic half-life are used to compute $p_{\text{intrinsic}}$ at C_{max} and the threshold concentration. The results (Table 2) show that all TCEs are predicted to have <50% intrinsic clearance at C_{max} , while most mAbs have >50% intrinsic clearance, consistent with the observed PK profiles (Figures 4-5). The authors also validate the trimeric binding expressions numerically (Figures 9-11) and the two-compartment effects (Figures 12-13). No formal statistical hypothesis testing or benchmark comparisons to alternative methods are performed; the validation is qualitative and illustrative.

Comparison to Alternatives

The work extends the classical TMDD framework of Mager & Jusko (2001) and the quasi-steady-state approximation of Gibiansky et al. (2008) by deriving an explicit algebraic expression for the fraction of clearance that is intrinsic ($p_{\text{intrinsic}}$). Compared to the in vivo potency concept (L_{50}) of Gabrielsson et al. (2018), the authors show that under the assumption $k_{\text{deg}} = k_e$, their threshold $L_{\text{free}} = k_D + k_I$ is equivalent to L_{50} . The key advantage is that $p_{\text{intrinsic}}$ and the threshold $L_{\text{free}} = (k_e/k_{\text{clear}}) \cdot R_{\text{tot}} - k'_D$ require only readily available parameters (k_{clear} , k_e , R_{tot} , k_D , k_{on}) and no numerical simulation. The method is simpler than full ODE-based TMDD models and provides immediate insight into clearance mechanism dominance. A limitation is that it does not capture the full multi-exponential ‘wobble’ region, but it correctly identifies the monoexponential regimes above and below that region.

Implementation Guidance

The method is implemented in Python (code provided in supplementary materials). To use the rule-of-thumb: (1) Estimate intrinsic clearance rate k_{clear} (e.g., from a high-dose phase where target is saturated). (2) Obtain target parameters: total receptor concentration R_{tot} , binding affinity k_D , on-rate k_{on} , and complex elimination rate k_e . (3) Compute $k'_D = k_D + k_e/k_{\text{on}}$. (4) Compute the threshold $L_{\text{thresh}} = (k_e/k_{\text{clear}}) * R_{\text{tot}} - k'_D$. If $L_{\text{thresh}} < 0$, clearance is always intrinsic-dominated. If $L_{\text{thresh}} > 0$, then for $L_{\text{free}} \gg L_{\text{thresh}}$ clearance is intrinsic-dominated, and for $L_{\text{free}} \ll L_{\text{thresh}}$ clearance is TMDD-dominated. The proportion of intrinsic clearance at any concentration is $p_{\text{intrinsic}} = 1/(1 + (k_e/k_{\text{clear}}) * (R_{\text{free}}/k'_D))$. Computational cost is negligible (algebraic evaluation).

☒ Key Findings

1. The proportion of clearance that is intrinsic ($p_{\text{intrinsic}}$) is given by a simple algebraic expression: $p_{\text{intrinsic}} = 1/(1 + (k_e/k_{\text{clear}}) * (R_{\text{free}}/k'_D))$.
 2. The transition from intrinsic-dominated to TMDD-dominated clearance occurs at $L_{\text{free}} = (k_e/k_{\text{clear}}) * R_{\text{tot}} - k'_D$. If this quantity is negative, TMDD never dominates.
 3. For T-cell engagers, even though target concentrations are low (e.g., CD3 ~ 0.1 nM), fast receptor turnover (k_e large) makes the threshold positive, explaining why TMDD dominates at pharmacologically active concentrations.
 4. Application to published PK data shows that all TCEs examined are predicted to have TMDD-driven clearance at C_{max} , while most monoclonal antibodies are intrinsic-driven, consistent with observed PK profiles.
 5. The expression provides a practical rule-of-thumb for selecting between allometric scaling (intrinsic clearance) and target-parameter-based scaling (TMDD) for cross-species PK translation.
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Strengths & Limitations

Strengths

- Derives a simple, closed-form algebraic expression for the proportion of intrinsic clearance, requiring no numerical simulation.
- Provides an intuitive threshold concentration that separates intrinsic-dominated from TMDD-dominated clearance.
- Explains why T-cell engagers can exhibit monoexponential PK that is nonetheless TMDD-driven, resolving a common confusion in the field.
- Validates the expressions against published PK data for multiple TCEs and mAbs, showing consistent classification.
- Includes extensions to two-compartment models and trimeric binding in supplementary materials.
- Code is provided for reproducibility.

Limitations (Acknowledged by Authors)

- Assumption that $k_{\text{deg}} = k_e$ (receptor degradation rate equals complex elimination rate) may not hold for all targets.

- Trimeric binding analysis depends on an unknown avidity factor (α).
- Finite k_{on} can cause divergence between theoretical and numerical $p_{\text{intrinsic}}$ at very low L_{free} (Appendix A.2).
- Parameter uncertainty in literature values for receptor densities, turnover rates, and binding affinities.
- Two-compartment effects can modify the apparent half-life but not the concentration thresholds.

Limitations (Expert Review)

- The derivation assumes constant total receptor concentration R_{tot} , but in reality target depletion (e.g., due to T-cell engagement) can occur, which would alter the thresholds over time.
- The method does not provide a formal statistical test to distinguish intrinsic vs. TMDD-driven clearance from observed PK data; it is a deterministic rule-of-thumb.
- The validation against published PK data is qualitative (visual inspection of plots) rather than quantitative (e.g., formal model comparison).
- The trimeric binding analysis in the appendix is preliminary and not fully integrated into the main results; the assumption that trimer elimination rate is the sum of dimer elimination rates may be unrealistic.

Generalizability

The core expressions are derived from a general TMDD model and should apply to any biologic that binds to a target with known parameters. The authors demonstrate applicability to both T-cell engagers (bispecifics) and monoclonal antibodies (including those with soluble targets). The extension to two-compartment models and trimeric binding in the appendices broadens the scope. However, the assumption $k_{\text{deg}} = k_{\text{e}}$ may not hold for all targets, and the trimeric analysis depends on an unknown avidity factor.

Key Equations

Total ligand dynamics under QSS

$$\frac{dL_{\text{tot}}}{dt} = - \left(k_{\text{clear}} + k_{\text{e}} \frac{R_{\text{free}}}{k'_{\text{D}}} \right) L_{\text{free}}$$

The quasi-steady-state approximation for the TMDD model, expressing total ligand dynamics in terms of free ligand and receptor concentrations.

Intrinsic clearance proportion

$$p_{\text{intrinsic}} = \frac{1}{1 + \frac{k_{\text{e}}}{k_{\text{clear}}} \frac{R_{\text{free}}}{k'_{\text{D}}}} = \frac{1}{1 + \frac{k_{\text{e}}}{k_{\text{clear}}} \frac{R_{\text{tot}}}{k'_{\text{D}} + L_{\text{free}}}}$$

The proportion of clearance that is due to intrinsic (non-specific) routes, derived from the total ligand equation.

Threshold for 50% intrinsic clearance

$$L_{\text{free}} \geq \frac{k_e}{k_{\text{clear}}} R_{\text{tot}} - k'_D$$

The threshold free ligand concentration at which intrinsic and TMDD-driven clearance are equal ($p_{\text{intrinsic}} = 0.5$).

QSS dissociation constant

$$k'_D = \frac{k_{\text{off}} + k_e}{k_{\text{on}}} = k_D + k_I$$

The effective dissociation constant under quasi-steady-state, accounting for complex elimination.

In vivo potency (L_{50})

$$L_{50} := \frac{k_{\text{deg}}}{k_e} \frac{k_{\text{off}} + k_e}{k_{\text{on}}} := \frac{k_{\text{deg}}}{k_e} (k_D + k_I)$$

The in vivo potency (L_{50}) from Gabrielsson et al., shown to be equivalent to $k_D + k_I$ when $k_{\text{deg}} = k_e$.

Figures & Tables

- **Figure 1:** Heatmaps showing the value of k_{clear} required to make intrinsic clearance 50% of total, as a function of L_{tot} , R_{tot} , k_D , and k_e .
 - *Significance:* Illustrates the intuitive parameter dependencies: higher k_e or lower k_D increases the k_{clear} needed for 50% intrinsic clearance, meaning TMDD dominates more easily.
- **Figure 2:** Contour plots of $L_{\text{free}}/L_{\text{tot}} = 0.5$, $C/R_{\text{tot}} = 0.5$, and $p_{\text{intrinsic}} = 0.5$ in parameter space (L_{tot} , k_D , $(k_e/k_{\text{clear}}) \cdot R_{\text{tot}}$).
 - *Significance:* Shows that $p_{\text{intrinsic}} = 0.5$ and $L_{\text{free}}/L_{\text{tot}} = 0.5$ are identical contours, confirming the direct relationship between free fraction and clearance mechanism.
- **Figure 3:** Simulated free ligand concentration vs. time for various parameter combinations, annotated with theoretical thresholds for clearance regime changes.
 - *Significance:* Validates that the derived thresholds (Eq. 7 and $L_{\text{free}} = k_D$) correctly predict the onset of TMDD-driven monoexponential decay and the ‘wobble’ region.
- **Figure 4:** Clinical PK data for several T-cell engagers (e.g., Epcoritamab, Glofitamab) with annotated theoretical thresholds.
 - *Significance:* Demonstrates that TCEs are predicted to be in a TMDD-driven regime at all observed concentrations, explaining their monoexponential PK despite low doses.
- **Figure 5:** Clinical PK data for several monoclonal antibodies (e.g., Anifrolumab, Canakinumab) with annotated theoretical thresholds.
 - *Significance:* Shows that most mAbs are predicted to have intrinsic-driven clearance, except Anifrolumab which shows a TMDD ‘wobble’ consistent with the derived threshold.

Code & Reproducibility Assessment

All simulation and analysis performed in Python 3.8.10 using scipy, numpy, matplotlib, pandas, sympy, and other standard libraries. Code to reproduce results is provided in supplementary materials with a README file. The repository is available upon request from the authors.

Supplementary Materials

The supplementary materials include: (A.1) Detailed analysis of monoexponential decay regimes for L_{free} under various conditions ($L_{\text{free}} \gg k'_D$, $L_{\text{free}} \ll k'_D$, etc.) with numerical confirmation in Figure 6. (A.2) Alternative results with finite k_{on} (Figures 7-8), showing that the QSS approximation may diverge at low L_{free} . (A.3) Trimeric binding analysis: derivation of dimer and trimer concentrations, a 'trimeric affinity' k_D^{trimer} , and trimer-driven half-life expressions, with numerical validation (Figures 9-11) and application to TCEs (Table 4). (A.4) Two-compartment effects: derivation of effective elimination rates for fast and slow inter-compartment transfer (Figures 12-13). (A.5) Detailed parameter values obtained from literature for all targets and drugs used in the analysis.

Future Directions

Future work should (1) relax the assumption $k_{\text{deg}} = k_e$ and derive expressions for the general case, (2) validate the trimeric binding results with experimental avidity data, (3) extend the framework to account for time-varying receptor concentrations (e.g., target depletion), and (4) develop a formal statistical test to distinguish intrinsic vs. TMDD-driven clearance from observed PK data.

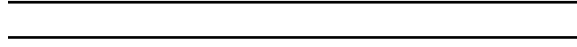
Expert Commentary

This paper fills an important practical gap: how to quickly diagnose whether a biologic's clearance is target-mediated without running full ODE simulations. The derived expression $p_{\text{intrinsic}}$ is elegant and intuitive. The key insight is that the threshold for TMDD dominance depends on the product $(k_e/k_{\text{clear}}) \cdot R_{\text{tot}}$, which explains why even low-abundance targets (like CD3 at ~0.1 nM) can drive TMDD if their turnover is fast (k_e large). The application to T-cell engagers is particularly valuable, as these drugs are often dosed at low levels where TMDD is expected but the PK appears monoexponential. The paper's main limitation is the assumption $k_{\text{deg}} = k_e$, which may not hold for all targets. Additionally, the trimeric binding analysis in the appendix is preliminary and depends on an unknown avidity factor. Nonetheless, the core result is robust and immediately useful for DMPK scientists.

Bottom Line

This paper provides a simple, algebraically derived rule-of-thumb (the intrinsic clearance proportion $p_{\text{intrinsic}}$ and the threshold $L_{\text{free}} = (k_e/k_{\text{clear}}) \cdot R_{\text{tot}} - k'_D$) that allows pharmacometricians to quickly determine whether a biologic's clearance is dominated by non-specific (intrinsic) routes or by target-mediated drug disposition (TMDD). For T-cell engagers and other

biologics dosed at low concentrations, the expression explains why monoexponential PK can still be TMDD-driven, and it guides the choice between allometric scaling (for intrinsic clearance) and target-parameter-based scaling (for TMDD).



☒ Figures

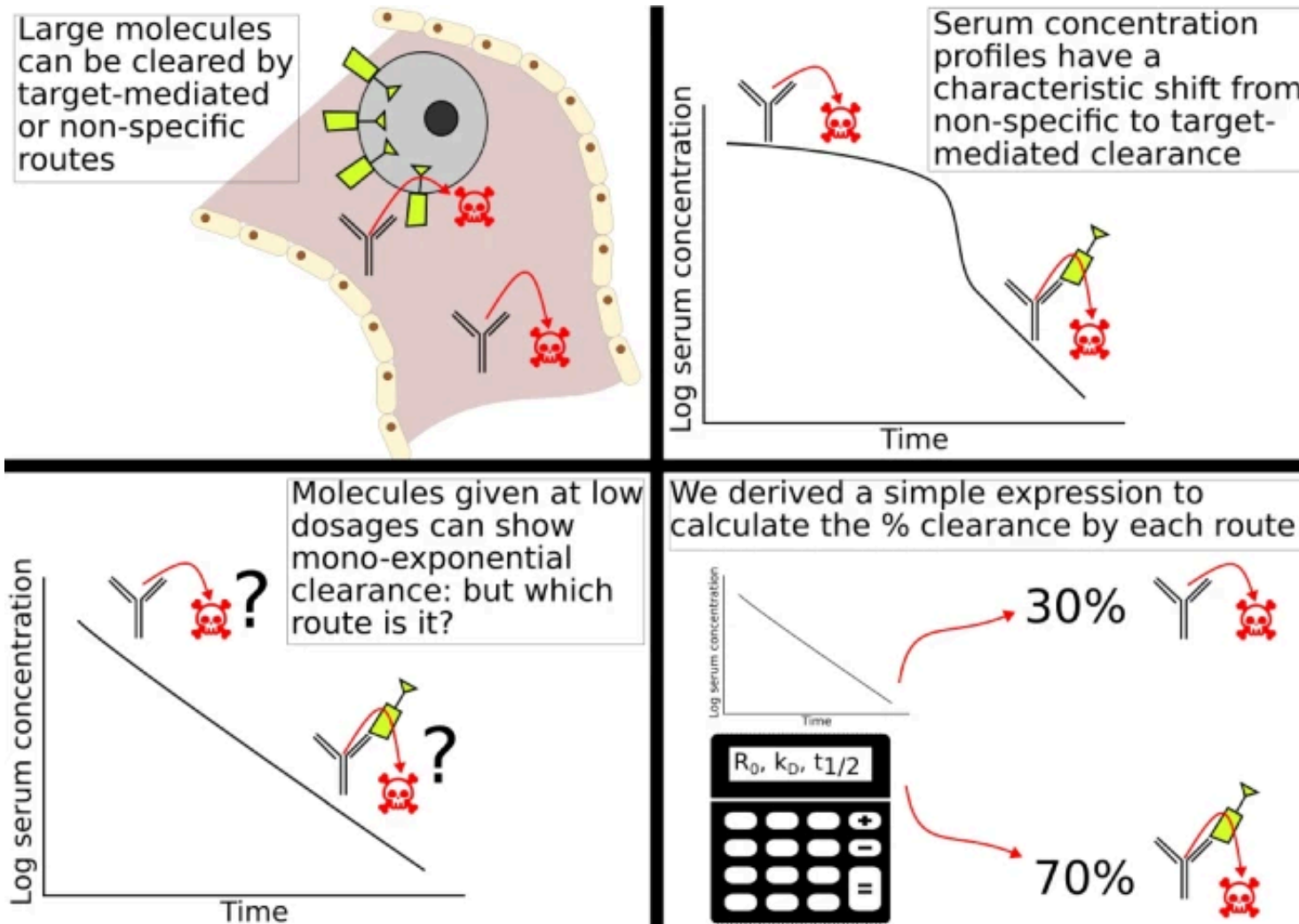


Figure 1: Figure 1

Typical use case for this paper’s results

- Obtain pharmacokinetic (PK) data from pre-clinical animal experiments for biologics or similar, which exhibits monoexponential clearance (“linear” on a log-scale)
- Cannot uniquely fit a PK model with both intrinsic and target-driven clearance – only one route can be quantified
- Using knowledge of target densities and turnover rates, use equation 5 to predict whether the molecule is in a TMDD-driven clearance regime
- Depending on if the clearance is intrinsic (allometric) and/or target-driven, use the most appropriate PK model and scaling technique(s) to predict clinical half-life.

Box 1: Intended application of this work to drug discovery. We provide a set of simple equations in Table 1 to aid model and scaling technique selection for DMPK and translational scientists.

Figure 2: Figure 2

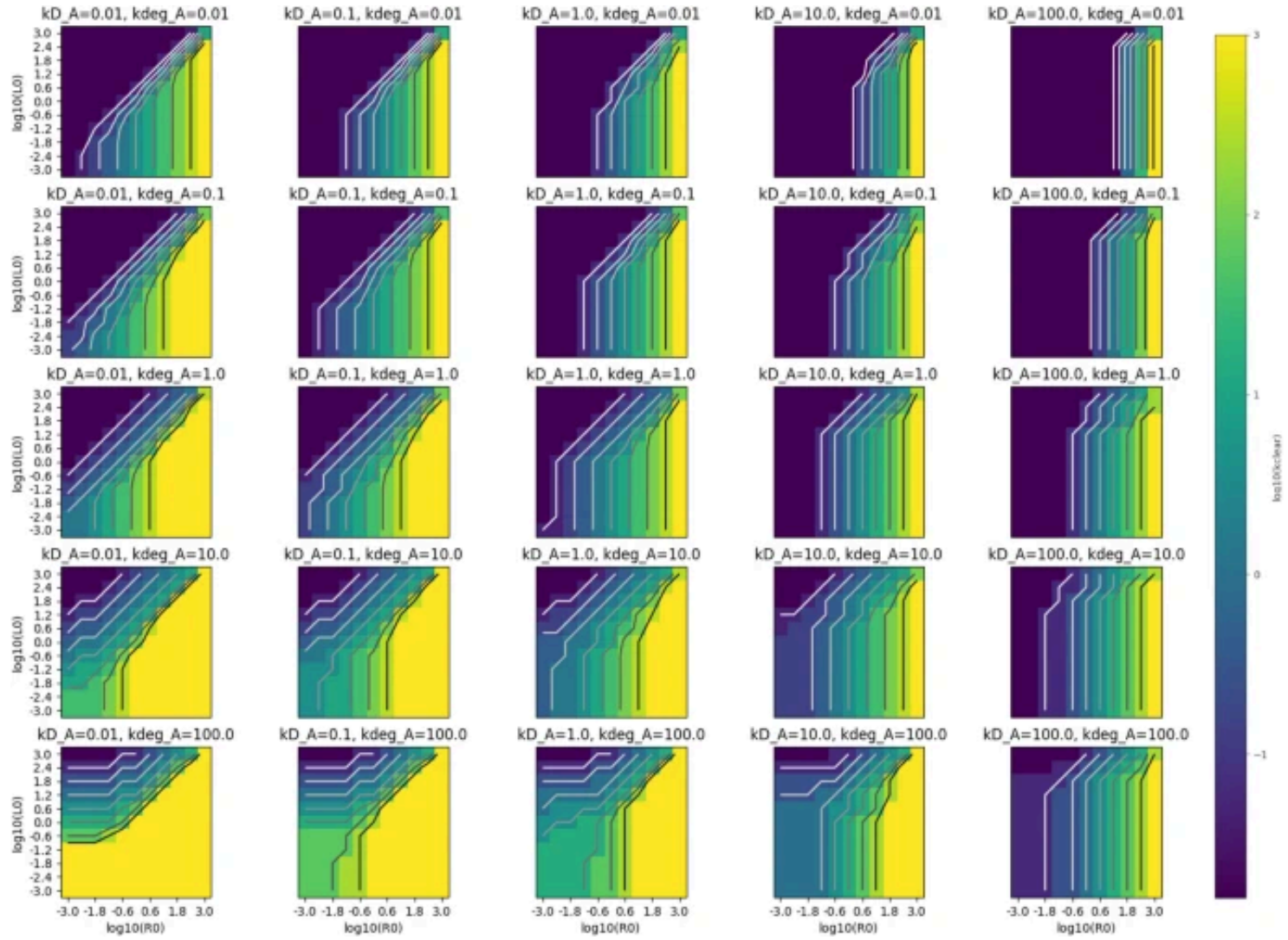


Figure 3: Figure 3

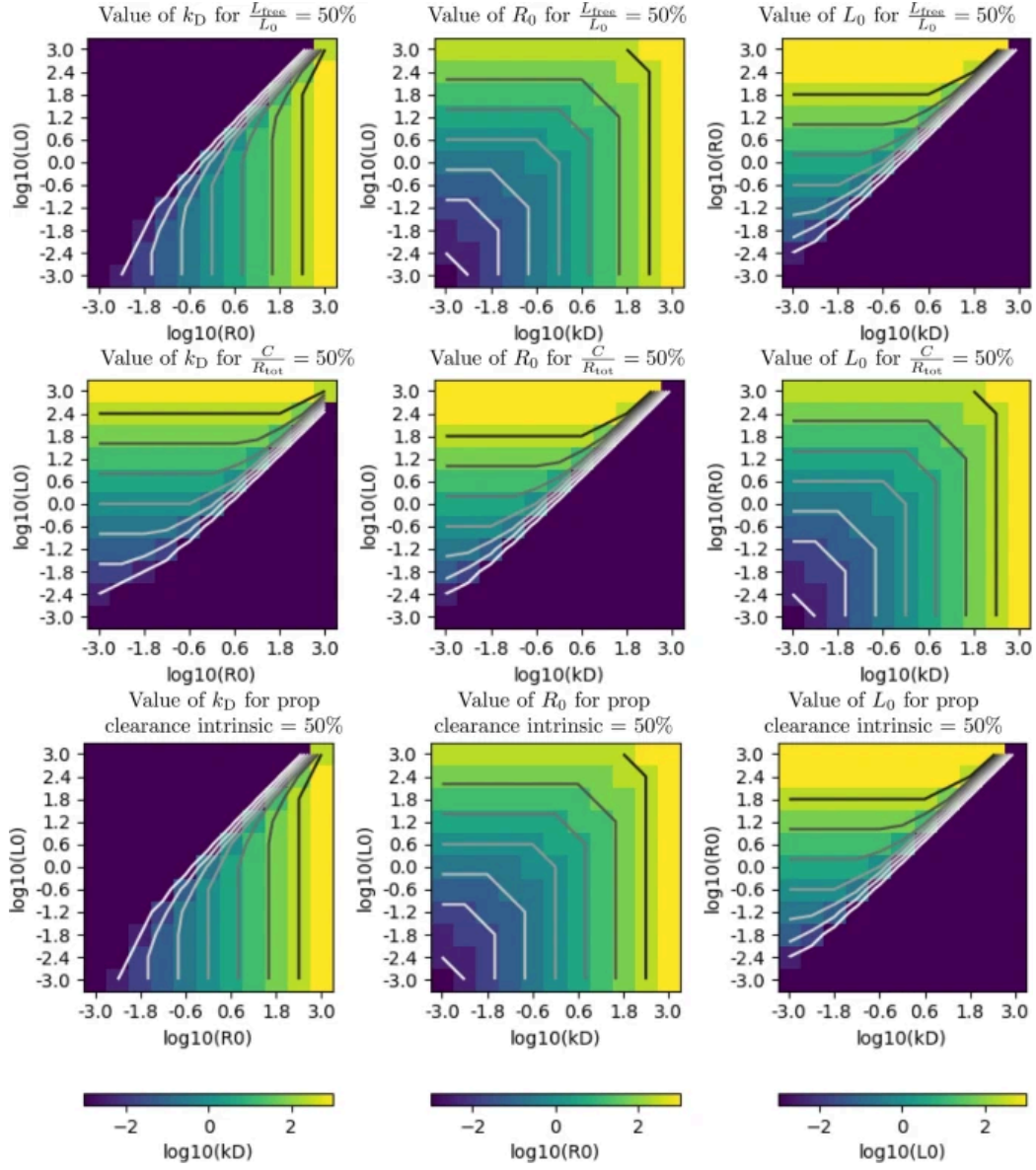


Figure 4: Figure 4

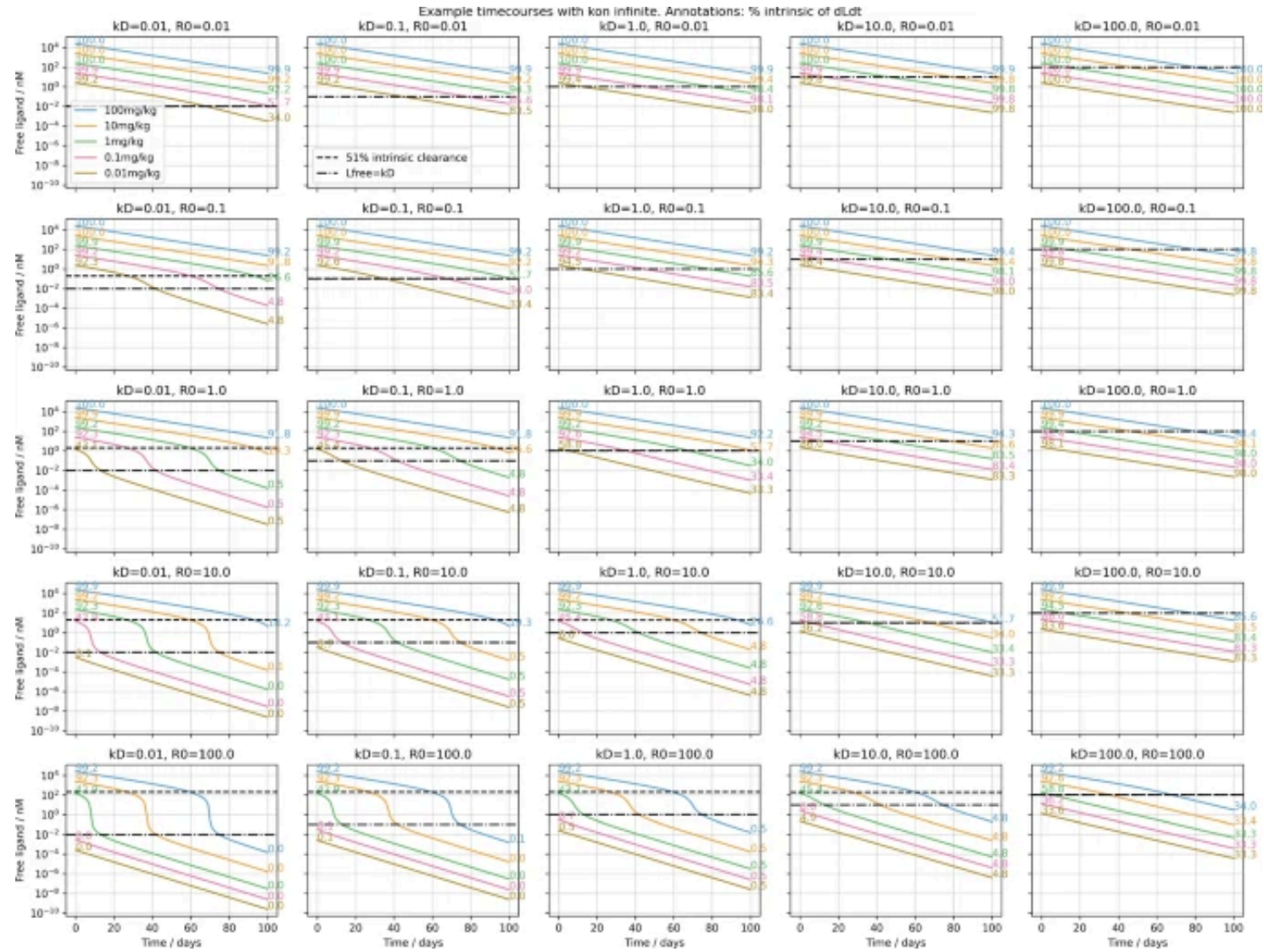


Figure 5: Figure 5

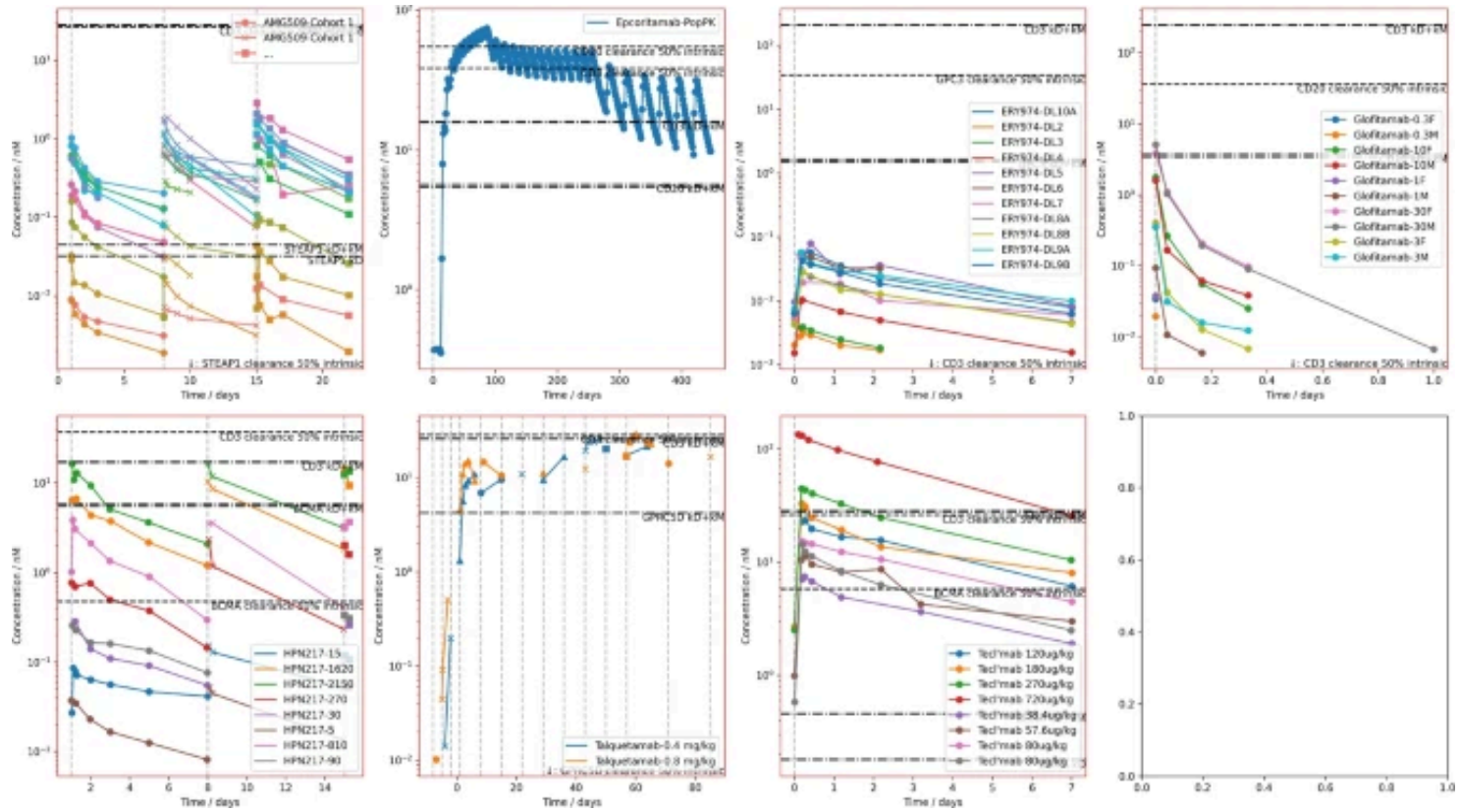


Figure 6: Figure 6

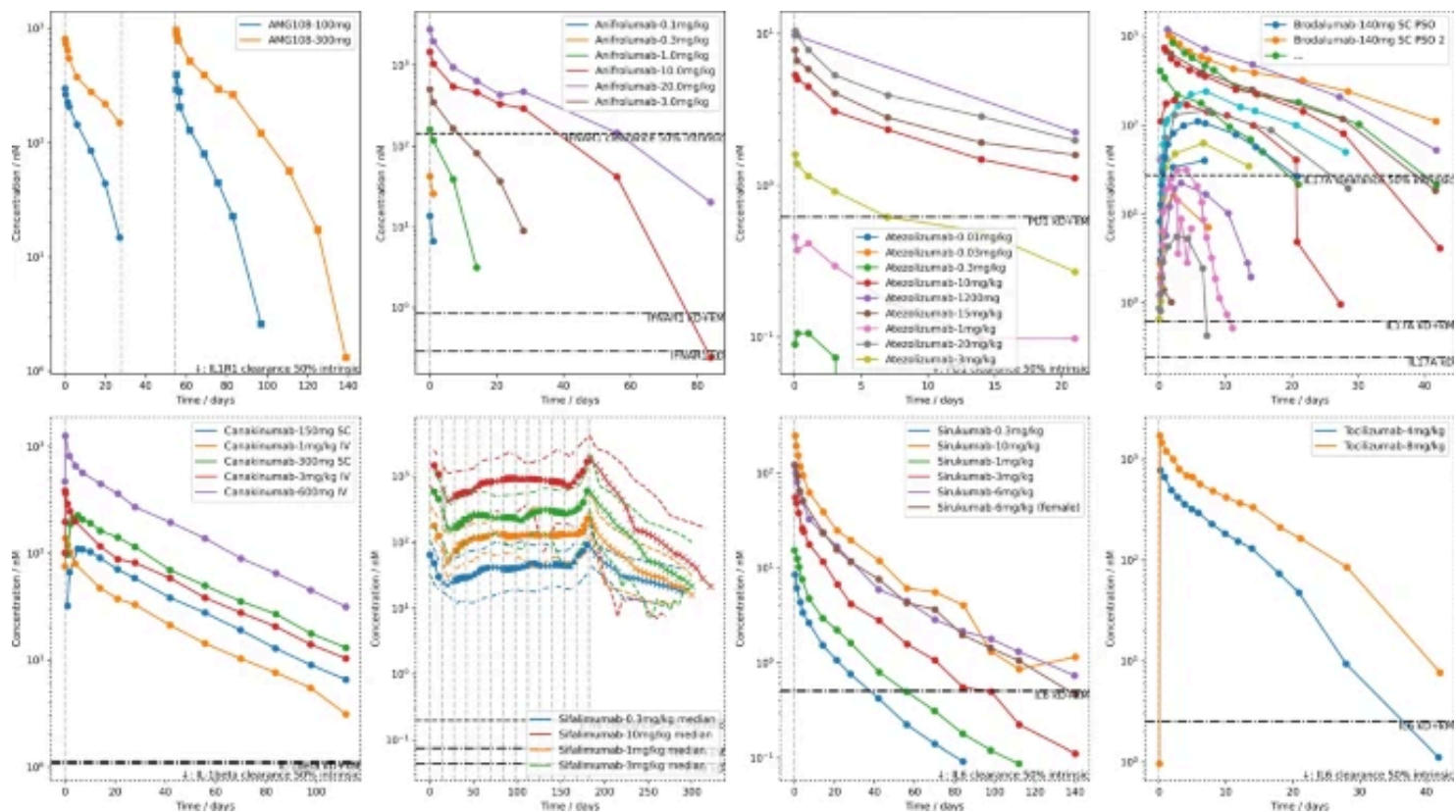


Figure 7: Figure 7

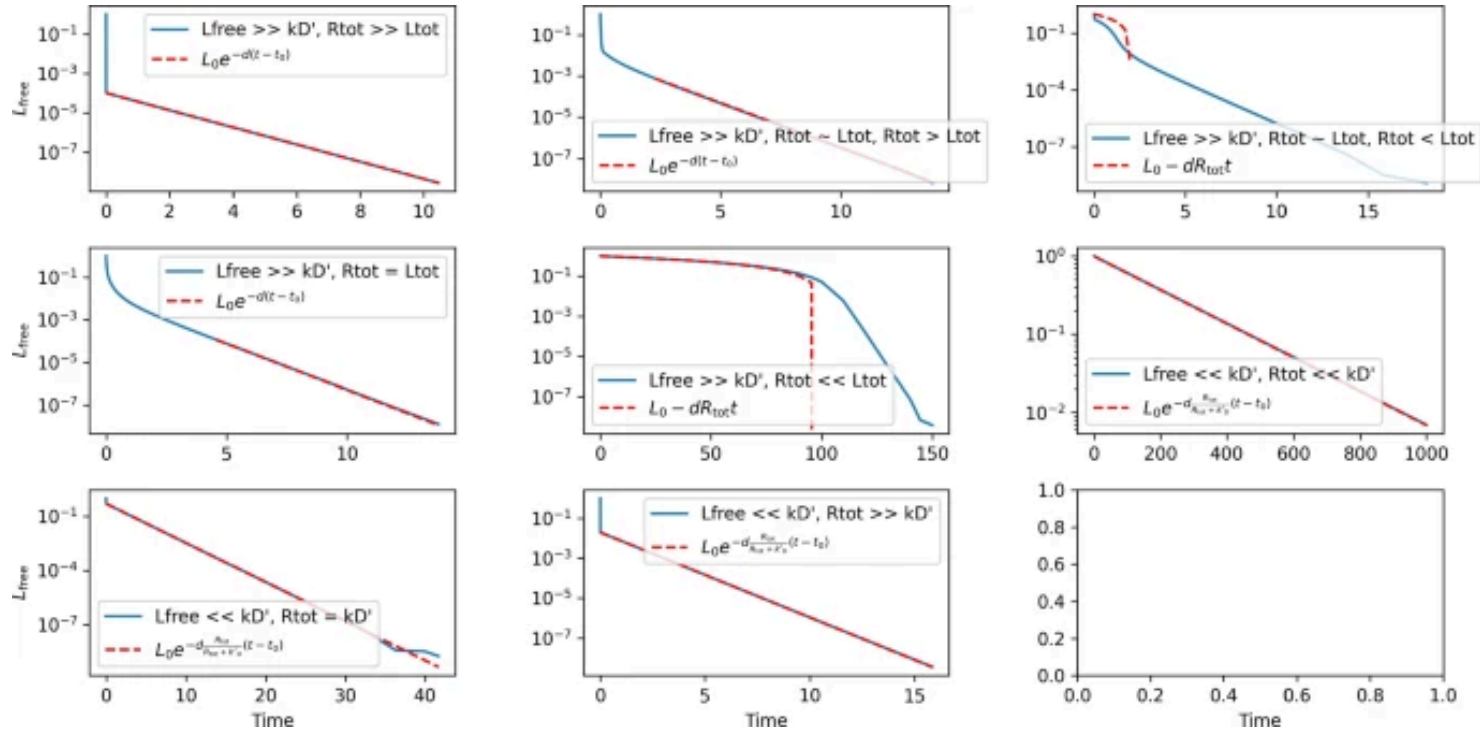


Figure 8: Figure 8

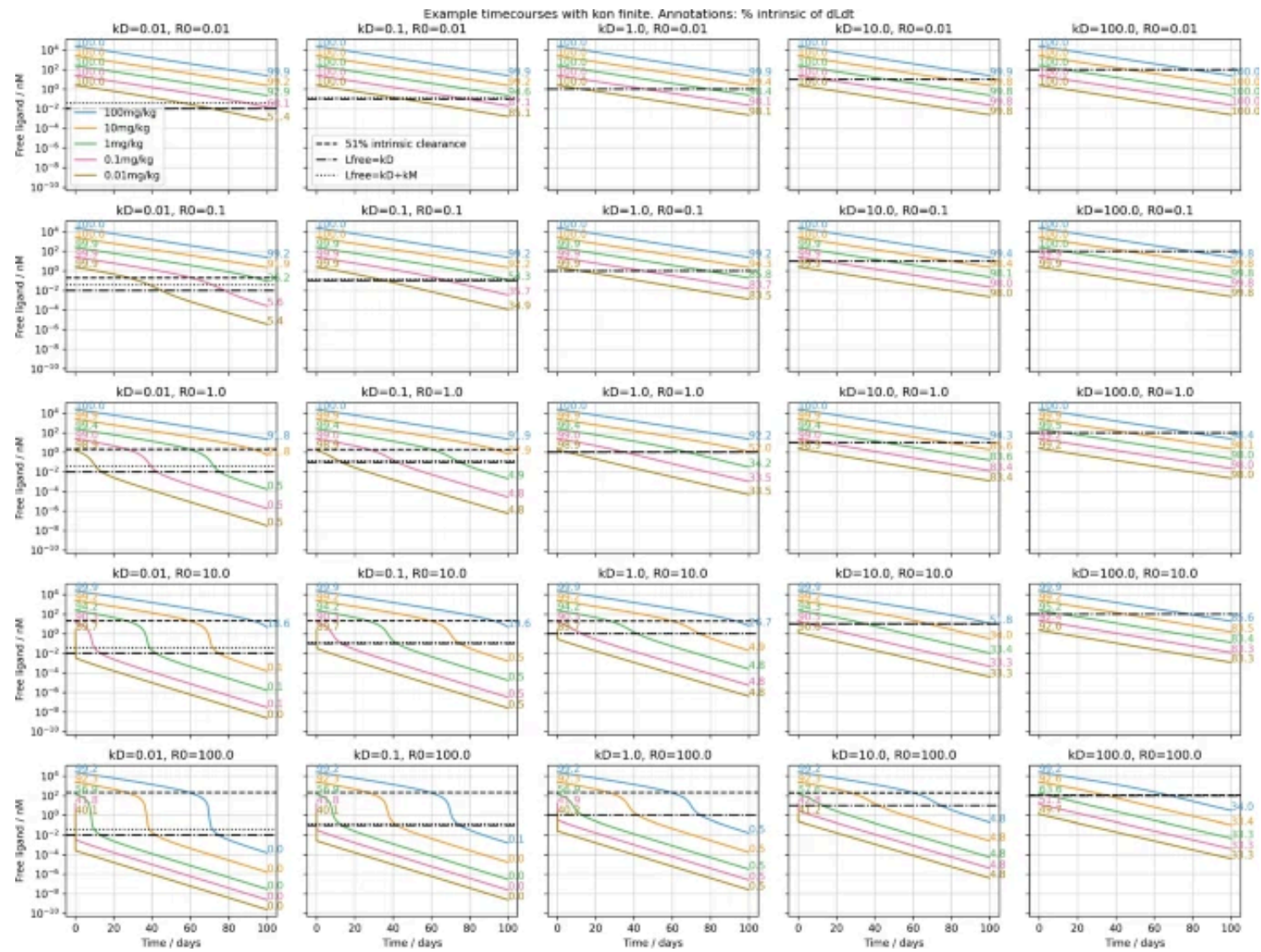


Figure 9: Figure 9

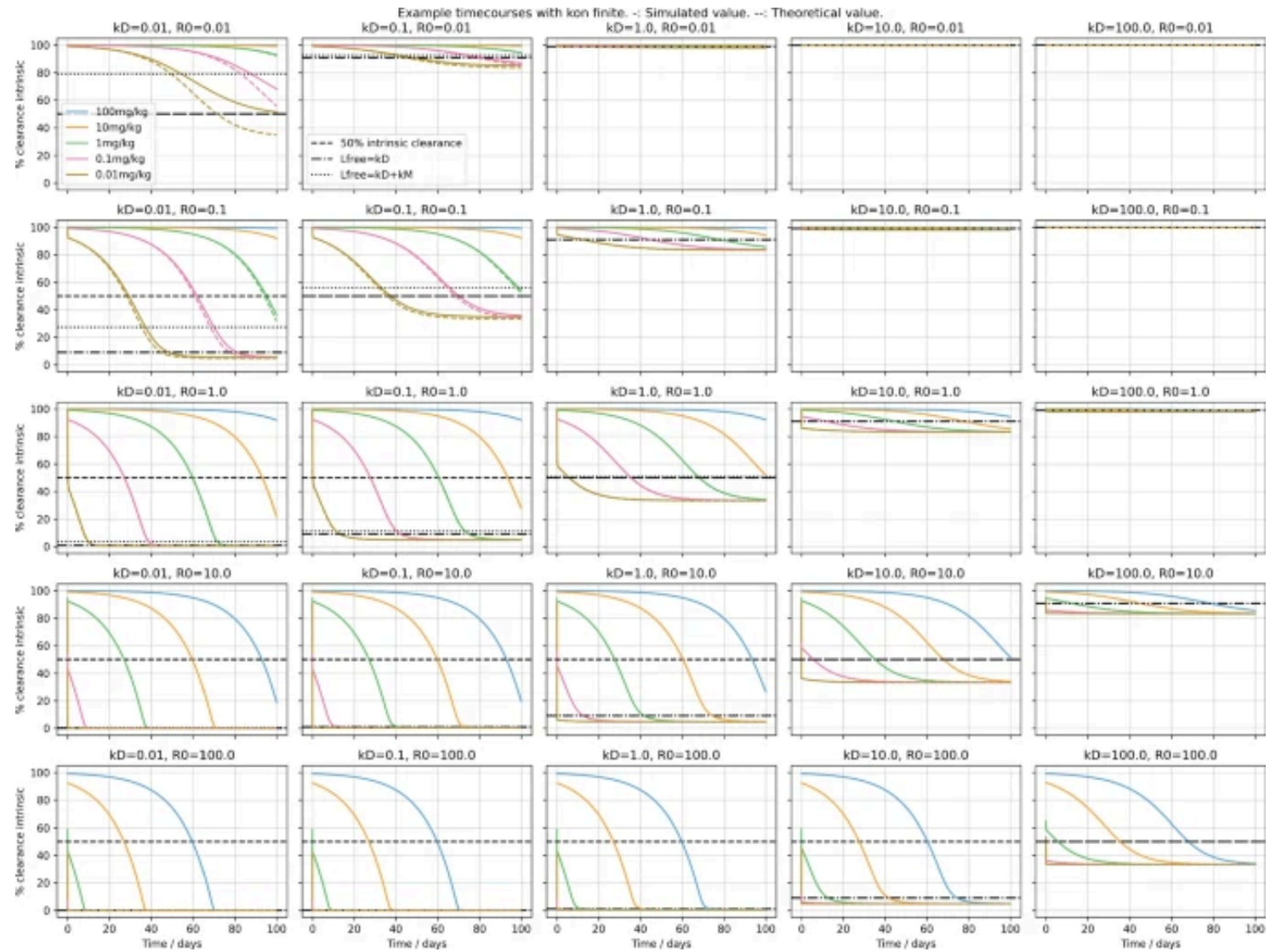


Figure 10: Figure 10

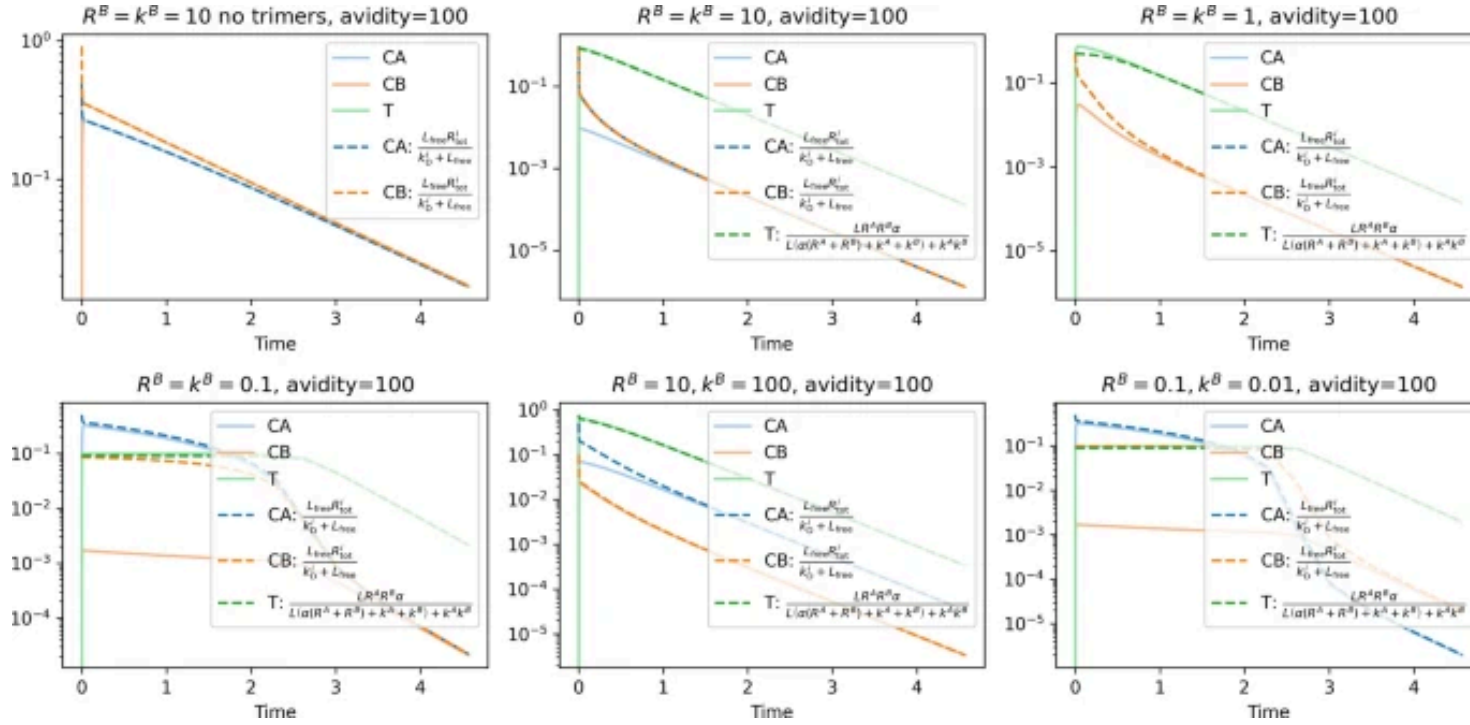


Figure 11: Figure 11

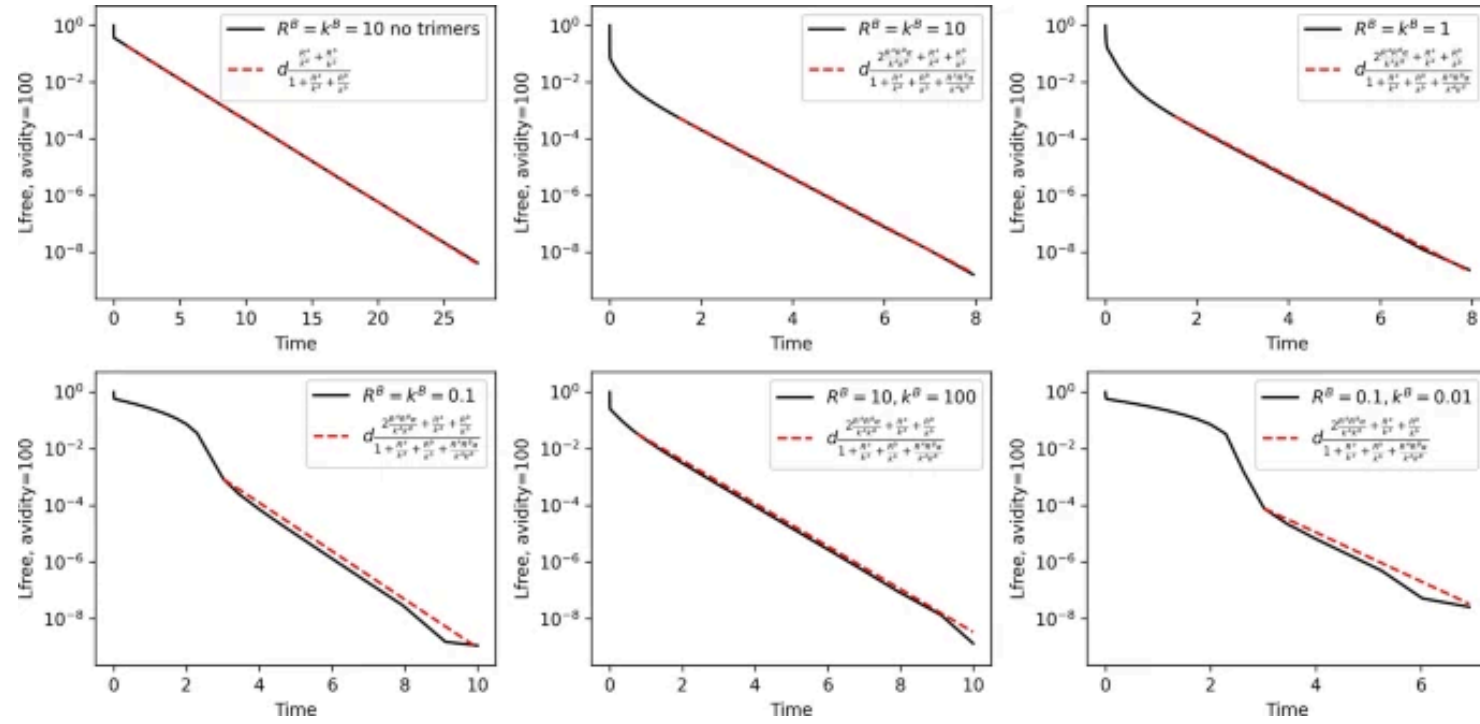


Figure 12: Figure 12

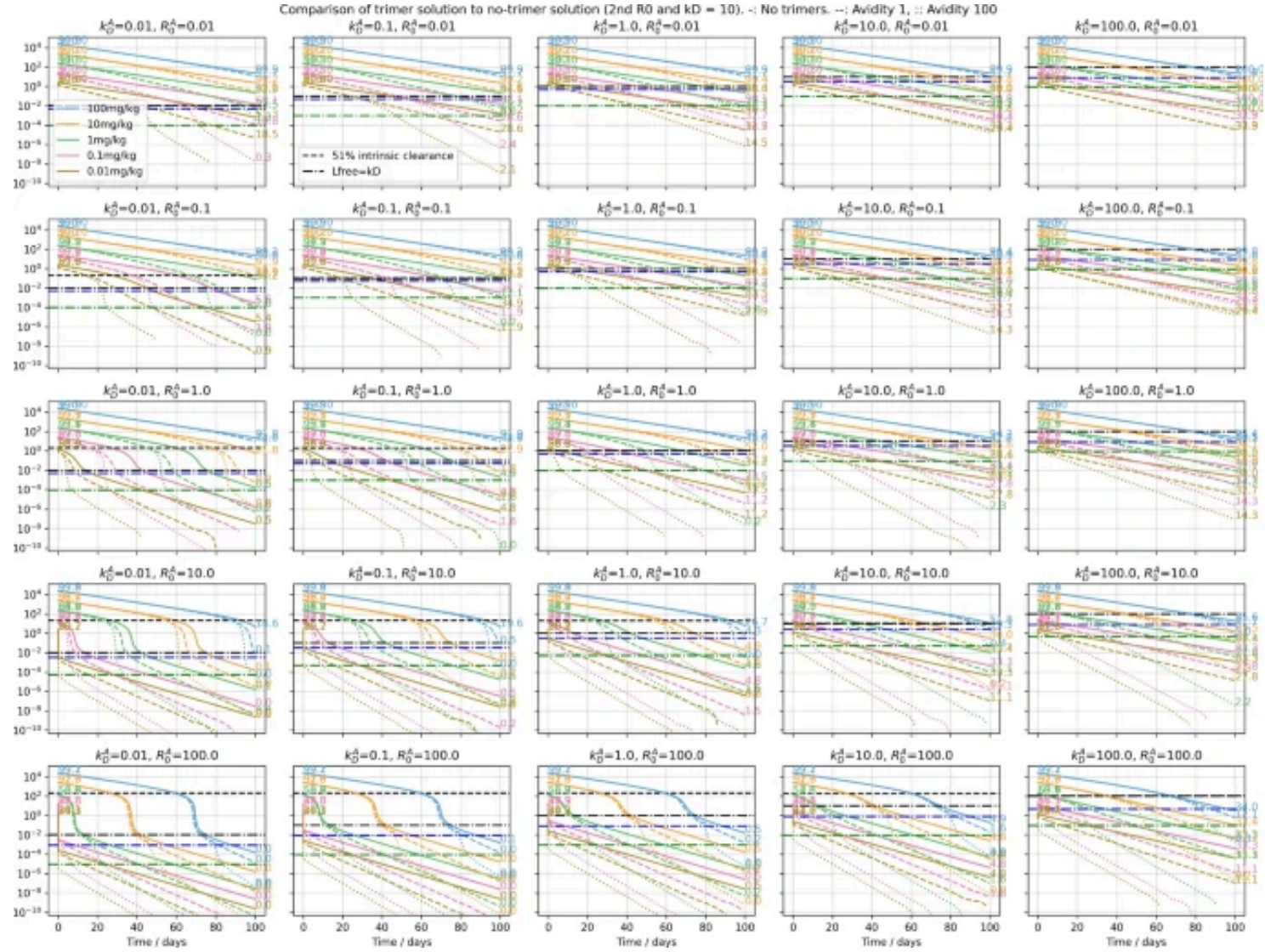


Figure 13: Figure 13

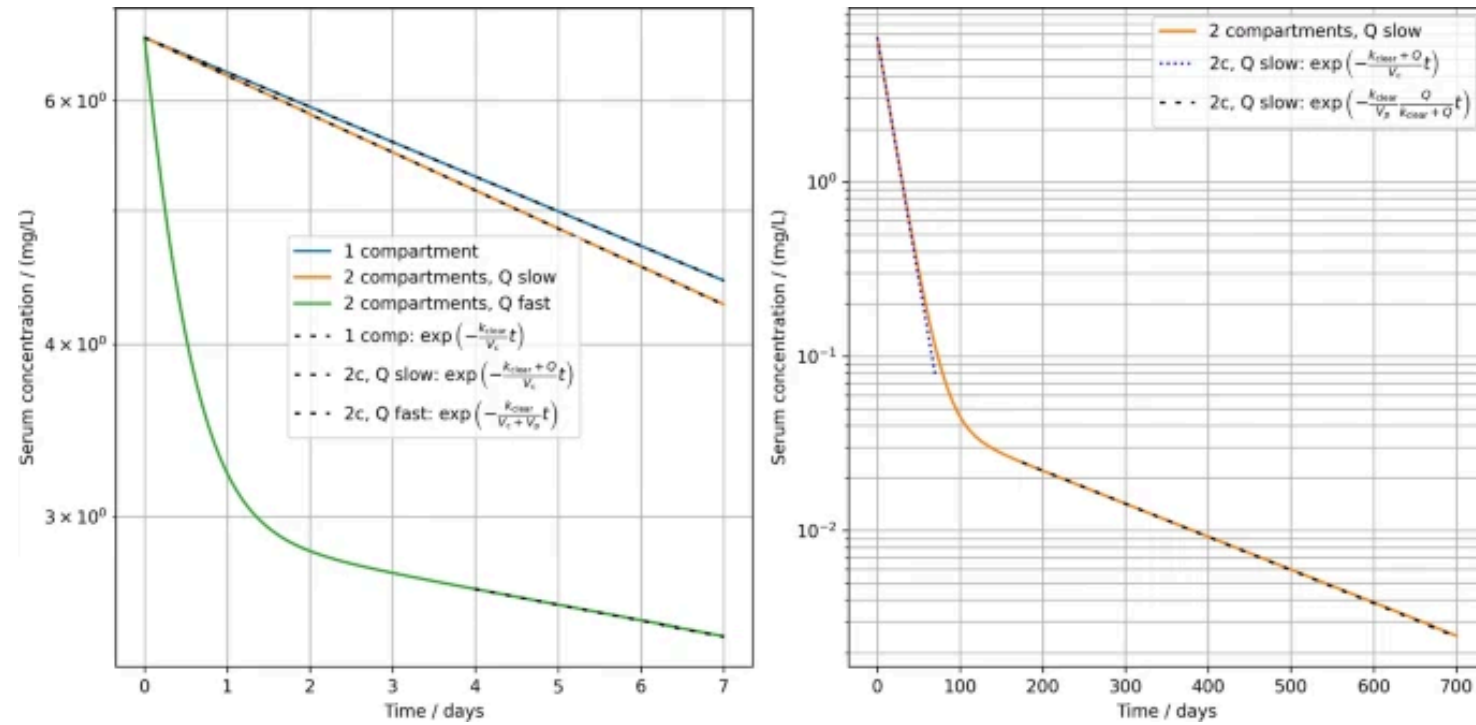


Figure 14: Figure 14

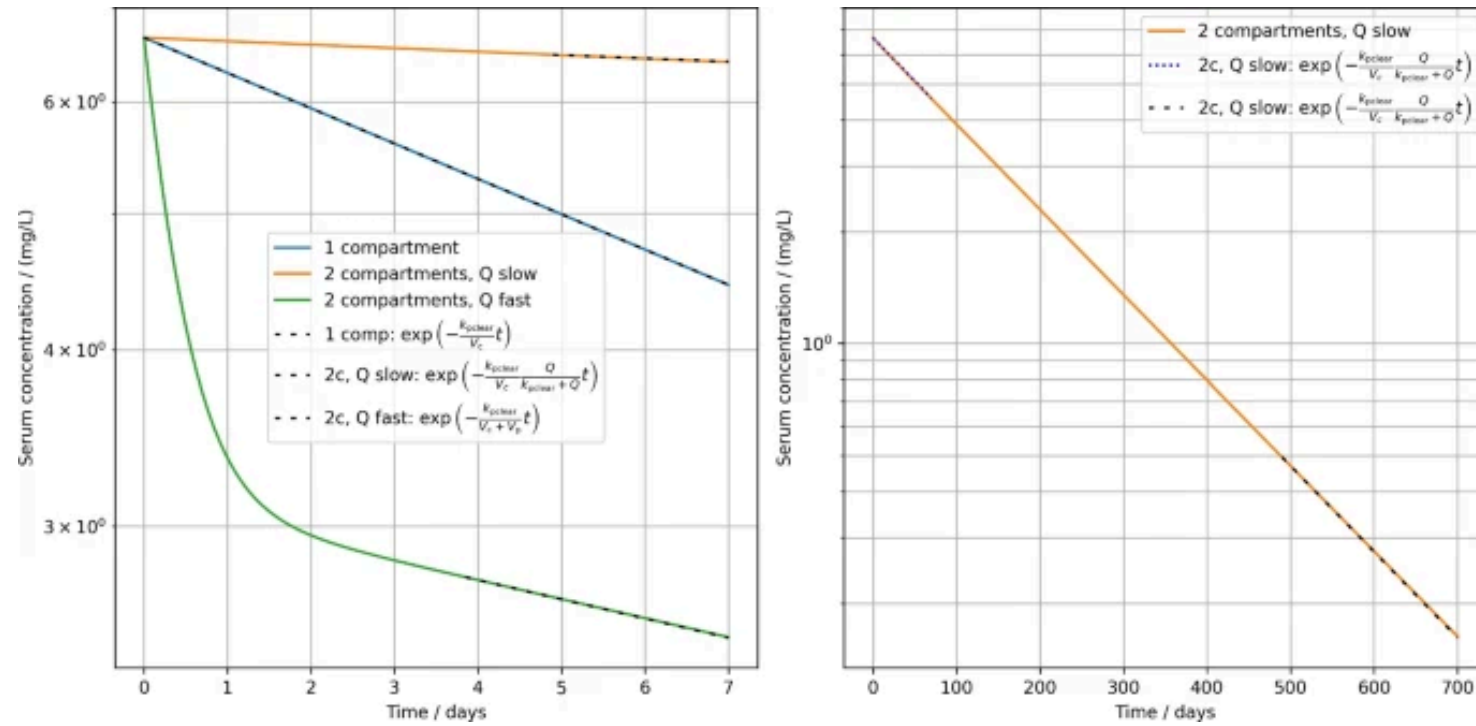


Figure 15: Figure 15